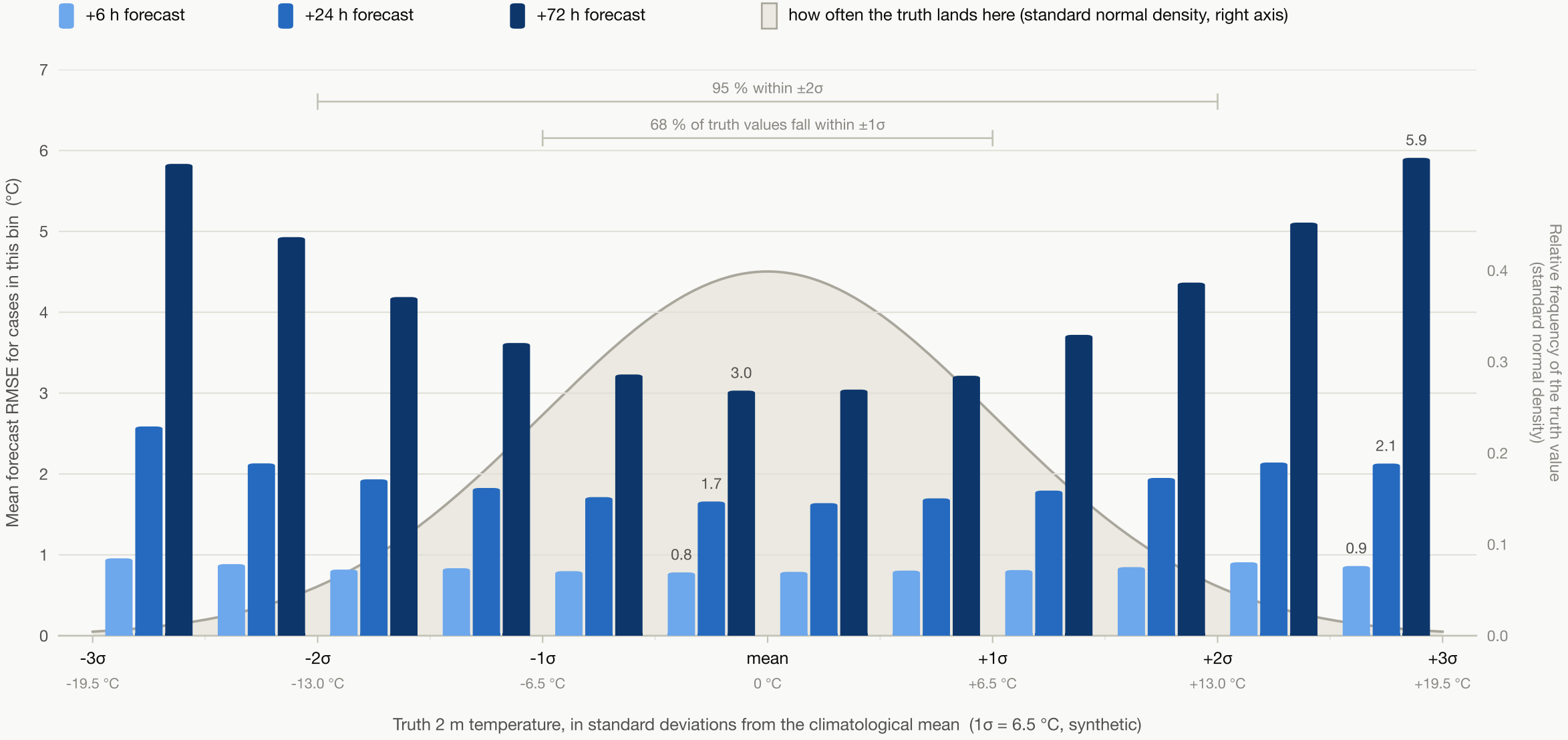
The further from normal the weather, the worse the forecast

Forecast RMSE for 2 m temperature, grouped by how unusual the observed value was. Errors are smallest near the climatological mean and grow toward both tails, and the growth steepens with lead time. The grey curve is the standard normal density: the tall tail bars describe events that are also the rarest.



Synthetic data, seed 1710. 20,000 verification cases with truth z-scores drawn from N(0, 1); 56 cases beyond ±3σ are not shown. Bin counts range from 87 (outer bins) to 3,859 (central bins).
Error model: forecast_z = r·truth_z + ε, ε ~ N(0, s^2), with (r, s) = (0.97, 0.12) at +6 h, (0.90, 0.25) at +24 h, (0.72, 0.45) at +72 h. The U shape is the r < 1 term: the model under-forecasts anomalies.